

Mohamed Zakaria Mohamed Ahmed

Software Engineer | Generative AI specialist

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SUMMARY

Software Engineer with hands-on experience designing and building AI-powered products. Proficient in Python and Node.js, with a strong foundation in RESTful API design and AI application integration. Experienced delivering solutions in fast startup environments. Seeking opportunities in Software Engineering, AI Engineering, and research or fellowship programs.

EDUCATION

BSc in Computer Science (Software Engineering)

INES-Ruhengeri, Rwanda — Expected Graduation: October 2026

Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Software Engineering, Artificial Intelligence

PROFESSIONAL EXPERIENCE

Software Engineer (July 2026 – Present)

Sokrab Group — Kigali, Rwanda

- Develop an AI-powered ERP agent that enables employees to interact with company ERP systems using natural language.
- Collaborate with operations teams to identify business processes and support the adoption of AI-powered solutions to improve efficiency and productivity.

AI Research & Engineering Intern (July 2026 – August 2026)

University of Parma — Parma, Italy

- Developed and evaluated an **ECG classification API** using a quantized TensorFlow Lite model, achieving **92.7% accuracy** on 1,000 test samples with ~18 ms single-image inference latency.
- Contributed to **research activities** under the Erasmus+ program, collaborating with researchers and gaining hands-on experience with AI model development and solar-energy systems.

Technologies: Python, FastAPI, TensorFlow Lite, INT8 Quantization, Machine Learning, REST APIs, PostgreSQL,

Software Engineer Intern (March 2026 – May 2026)

Digital Umuganda — Kigali, Rwanda

- Developed an AI-powered mobile app that combines vision-language models, OCR, and Kinyarwanda translation/TTS to provide real-time scene descriptions and text reading for people with visual disabilities. Built and orchestrated the end-to-end AI pipeline on edge hardware, integrating lightweight quantized models for scene understanding, translation, and Kinyarwanda speech generation.
- Collaborated with a dynamic team at Digital Umuganda, a startup focused on innovative tech solutions in Rwanda.

PROJECTS

Odoo AI Agent

[Demo Video](#)

- Built a conversational AI layer for Odoo that enables employees to query ERP data and perform business operations using natural language.
- Integrated Odoo MCP with a hybrid ORM + read-only SQL architecture, allowing the agent to execute Odoo-native actions while handling complex cross-model analytics.

Technologies: Fastapi, PostgreSQL, MCP, Vue.js, Docker, LLMs, langchain

VisionTTS

github.com/zakk-io/VisionTTS | [Demo Video](#)

- Built a real-time assistive application combining computer vision (VLM) and speech synthesis (TTS) for visually impaired users.
- Designed for edge deployment under hardware-constrained conditions, achieving accessible performance on low-resource devices.

Technologies: Flask, VLM, LLM, edge deployment.

AI-Powered Recruitment Platform (RAG-Based Candidate Screening)

github.com/zakk-io/Umurava-AI-Hackathon | [Demo Video](#)

- Designed and implemented an end-to-end AI recruitment system featuring automated CV screening, candidate evaluation, and AI-generated shortlist rankings.
- Built a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB vector databases and Gemini API, enabling context-aware querying across candidate documents.

Technologies: Node.js, Express.js, FastAPI, Vue.js, MongoDB, ChromaDB, WebSockets, Gemini API.

AI Pull Request Review Bot (CI/CD Automation)

github.com/zakk-io/pull_request_bot | [Demo Video](#)

- Developed an automated code review tool that performs static bug detection and security vulnerability analysis on pull requests.
- Integrated the bot into GitHub Actions CI/CD pipelines, reducing manual code review time and improving software quality assurance.

Technologies: GitHub Actions, CI/CD, Github api, gemini api

Real-Time Collaborative Code Editor

github.com/zakk-io/CoddyGame-api | [Demo Video](#)

- Architected a real-time collaborative coding environment supporting multiple concurrent users with low-latency synchronization.
- Implemented secure authentication using JWT and OAuth2 with role-based access control (RBAC) for session and permission management.

Technologies: Node.js, Vue.js, MongoDB, WebSockets.

TECHNICAL SKILLS

- **Programming Languages:** Python, Node.js
- **Frameworks & Libraries:** Express.js, Flask, FastAPI, Vue.js
- **AI:** Large Language Models (LLMs), Vision-Language Models (VLMs), OpenAI API Agents SDK, google ADK, langchain
- **Databases & Storage:** MongoDB (Mongoose), postgresql, ChromaDB (Vector Database)
- **Security & Auth:** JWT, OAuth2, Role-Based Access Control (RBAC), owasp top 10
- **Tools & Workflow:** Git, GitHub, Postman

ACHIEVEMENTS & COMMUNITY

- **AI Agents Fundamentals — Hugging Face** | April 2026
Completed training in AI agents, tool use, and multi-agent orchestration.
- **6th Place, Top 10 Startups — Code for Impact 2025, Mastery Hub Rwanda** | 2025
Developed and optimized a **Kinyarwanda text-to-speech API** for improved inference performance and production integration.
- **Appreciation Certificate — Sudanese Students Online Summit** | February 2026
Recognized for delivering a software engineering session to an sudanes student audience in Rwanda.
- **Workshop Facilitator — Career Development & Opportunities, SSOS Rwanda** | August 2026
Guided university students on career planning, scholarships, internships, and international opportunities.
- **Workshop Facilitator — Generative AI & Technology Fundamentals** | July 2026
Introduced high school students to AI, Generative AI, and AI agents through practical, beginner-friendly explanations.